Arkobrato Gupta

gupta.1161@osu.edu | +1-616-227-8780 | arkobrato.github.io
Google Scholar | arkobrato-gupta | ORCID: 0000-0001-5026-3085

EDUCATION

The Ohio State University — **Columbus, OH, USA**
- *Ph.D. in Biostatistics, GPA: 3.88/4.0* — *Aug 2021 - Aug 2027 (expected)*
- Ph.D. Candidate since Jan 2026
- Advisor: Dr. Dongjun Chung
- Dissertation: Statistical Frameworks for Integrating Multi-Omics & Multi-modal Data
- Coursework: Advanced Survival Analysis, Statistical Genetics, Causal Inference, Bayesian Inference, MCMC, Bioinformatics & Machine Learning in Omics

The Ohio State University — **Columbus, OH, USA**
- *M.S. in Statistics* — *Aug 2021 - Dec 2025*

Hindu College, University of Delhi — **New Delhi, India**
- *M.S. in Statistics* — *Jul 2017 - Jun 2019*

St. Xavier's College, University of Calcutta — **Kolkata, India**
- *B.S. (Hons.) in Statistics* — *Jul 2014 - Jun 2017*
- Supervisor: Durba Bhattacharya
- Thesis: Application of Bayesian hierarchical modeling in astro-statistics
- Fit a Bayesian hierarchical Poisson change-point model to Chandra X-ray flare data from young stars using hybrid Gibbs/Metropolis-Hastings MCMC in R

RESEARCH EXPERIENCE

The Ohio State University Comprehensive Cancer Center — **Columbus, OH, USA**
Graduate Research Associate — *May 2022 - May 2026*
Pelotonia Institute for Immuno-Oncology

Dissertation research
- **SRT:** Statistical framework for integrating spatial TCR-seq and spatial transcriptomics
  - Combines earth-mover's distance with Gaussian-mixture FDR control
  - Identified 878 genes coupled with T-cell clonal diversity in glioblastoma, enabling biomarker discovery in the tumor microenvironment
- **Opioid use disorder** (with multi-site collaborators)
  - Developed Bayesian stochastic block model clustering framework for 900+ genetically diverse rats
  - Classified heroin-use-disorder vulnerable, intermediate, and resilient phenotypes and revealed sex-specific subgroups within the vulnerable phenotype
  - Extending the framework to link behavioral subgroups with brain-region gene expression
- **SCOPE:** R Shiny app implementing this clustering framework
  - Modularity-initialized Gibbs sampling with posterior uncertainty quantification
  - Adopted by 5+ labs in the USA, Italy, and the UK
- **graph-GPA 2.0** (co-first author; *Frontiers in Genetics*, 2023)
  - Co-developed a Bayesian graphical model integrating GWAS summary statistics across multiple diseases with functional annotation data
  - Improved detection of disease-associated variants and estimation of genetic relationships among diseases
  - Applied to five autoimmune diseases and five psychiatric disorders

Collaborative projects
- **DNA methylation** (co-first author; *Scientific Reports*, 2025)
  - Co-led a genome-wide EWAS (Illumina EPIC; 802,978 CpGs; $n = 324$ postmenopausal women)

  - Identified 736 differentially methylated loci between Native Hawaiian and White women
  - Loci associated with liver disease pathways (61% of genes), DASH diet adherence, and adiponectin
- **Epigenetic aging in lung cancer** (Beating Lung Cancer in Ohio study, 2025)
  - Estimated epigenetic age and age acceleration (Horvath, PhenoAge, GrimAge) from EPIC methylation data in advanced NSCLC
  - Showed age-acceleration estimates were robust to methylation QC (paired t-tests, correlation analysis)
- **Comparative oncology** (*Cancers*, 2025)
  - Performed statistical analyses of clinical, histopathologic, and gene expression data from 75 dogs with soft tissue sarcoma
  - Higher FOXP3$^+$ regulatory T-cell infiltration was associated with shorter progression-free intervals
  - Tumors resembled human undifferentiated pleomorphic sarcoma, supporting canine sarcoma as a preclinical model
- **Glioblastoma myeloid landscape** (supported a grant application, 2025)
  - Analyzed public glioblastoma scRNA-seq data (100,000+ cells) to annotate six myeloid populations using Seurat, ScType, SingleR, and gene-signature scoring
  - Profiled immune checkpoint ligand expression (e.g., VISTA, PD-L1, TIM-3) across myeloid subsets

Indian Institute of Management (IIM) Ahmedabad **Ahmedabad, India**
Research Intern *Aug - Sep 2018*
- Collaborated with a multidisciplinary team to quantify public support for nuclear energy using survey data and regression analysis

National Sample Survey Office (NSSO), Govt. of India **Kolkata, India**
Research Intern *Jun - Jul 2018*
- Analyzed national sample survey data using logistic regression and survey methods to identify determinants of catastrophic health expenditure in India. Presented findings to the Ministry

INDUSTRY EXPERIENCE

Abbott Nutrition **Columbus, OH, USA**
Biostatistics Intern *Summers 2025 & 2026*
- **2025:**
  - Analyzed a propensity-matched NHANES cohort (n = 6,508; 2011–2018): adults consuming $\geq$1.2 g/kg/day protein had 1.2% greater lean soft tissue ($p < 0.001$), independent of physical activity, supporting protein guidelines for healthy aging
  - Owned the full analysis lifecycle: cohort definition, statistical method selection, confounder adjustment, and inference. Communicated results to scientific stakeholders and presented to leadership
  - Accepted for poster presentation at ICFSR 2026 (Washington, DC)
- **2026:**
  - Conducted an individual participant data (IPD) meta-analysis of 14 clinical trials using both two-stage and one-stage (mixed-effects) approaches, and explored mixture models to capture heterogeneity in intervention effects on key health outcomes
  - Communicated results to scientific stakeholders and presented to leadership

Novartis Healthcare Pvt. Ltd. **Hyderabad, India**
Statistical Programmer *Aug 2019 - Jul 2021*
- Programmed SDTM and ADaM datasets and tables, listings, and figures (TFLs) in SAS and R to CDISC standards for three clinical trials, including the pivotal Phase III study of Kisqali (ribociclib) in breast cancer
- Provided statistical programming support for early-phase studies with the Novartis Institutes for BioMedical Research (NIBR)
- Implemented the statistical methods specified in statistical analysis plans (SAPs), working with trial statisticians on analysis specifications and outputs
- Performed independent double programming and quality-control validation of datasets and outputs to ensure accuracy and submission readiness
- Built reporting databases supporting regulatory submissions across solid-tumor oncology and respiratory therapeutic areas

PUBLICATIONS

Peer-Reviewed Papers

∗ Joint first authors

- Song, M. A.*, **Gupta, A.***, Park, S. Y., Kim, J., Archer, K. J., Tiirikainen, M., Franke, A., Wilkens, L. R., Le Marchand, L., Chung, D., & Lim, U. (2025). Differential blood DNA methylation loci between Native Hawaiian and White women are associated with dietary patterns and metabolic biomarkers. *Scientific Reports*, 15, 43191. doi:10.1038/s41598-025-25253-4
- Kuhn, B. N., Cannella, N., Crow, A. D., Lunerti, V., **Gupta, A.**, Walterhouse, S. J., …, Chung, D., Ciccocioppo, R., & Kalivas, P. W. (2025). Distinct behavioral profiles and neuronal correlates of heroin vulnerability versus resiliency in a multi-symptomatic model of heroin use disorder in rats. *American Journal of Psychiatry*, 182(2), 198–208. doi:10.1176/appi.ajp.20230623
- Kuhn, B. N., Cannella, N., Chitre, A. S., …, **Gupta, A.**, …, Chung, D., … & Palmer, A. A. (2025). Genome-wide association study reveals multiple loci for nociception and opioid consumption behaviors associated with heroin vulnerability in outbred rats. *Molecular Psychiatry*, 30(8), 3363–3375. doi:10.1038/s41380-025-02922-4
- Hayburn, R., Chung, D., **Gupta, A.**, Sills, S., Wu, J. D., Ambrus, A., Jing, W., Ng, J., Penaloza-MacMaster, P., Selmek, A., Pollack, S. M., & Bracha, S. (2025). The immune landscape of canine soft tissue sarcomas as a model for human soft tissue sarcomas. *Cancers*, 17(23), 3860. doi:10.3390/cancers17233860
- Deng, Q.*, **Gupta, A.***, Jeon, H.*, Nam, J. H., Yilmaz, A. S., Chang, W., Pietrzak, M., Li, L., Kim, H. J., & Chung, D. (2023). graph-GPA 2.0: improving multi-disease genetic analysis with integration of functional annotation data. *Frontiers in Genetics*, 14. doi:10.3389/fgene.2023.1079198
- Jeon, H., Xie, J., Jeon, Y., Jung, K. J., **Gupta, A.**, Chang, W., & Chung, D. (2023). Statistical power analysis for designing bulk, single-cell, and spatial transcriptomics experiments: review, tutorial, and perspectives. *Biomolecules*, 13(2), 221. doi:10.3390/biom13020221

Preprints

- Shin, J., Yoo, J., Cho, Y., **Gupta, A.**, Mao, P.-H., Thakkar, K., Eddy, T. A., Chung, K. J., Kim, J., Jeon, H., Xie, J., & Chung, D. (2026). Circular data analysis for spatial omics. *bioRxiv*. doi:10.64898/2026.09.29.755467
- Walterhouse, S. J., Crow, A. D., Dereschewitz, E., Hopkins, J. L., **Gupta, A.**, Hodebourg, R., Chung, D., Kalivas, P. W., Garcia-Keller, C., & Kuhn, B. N. (2026). Attenuation of nucleus accumbens dendritic spine density following protracted abstinence in heroin vulnerable rats. *Preprint*.
- Nagy, G., Presley, R., Dhar, S., Jeon, H., Moreira, L., Tull, O., **Gupta, A.**, Chung, D., Tsichlis, P. N., & Parvin, J. D. (2024). Identification of TAP2 protein variants resistant to inhibition by the HSV1 ICP47 protein. *bioRxiv*. doi:10.1101/2024.11.18.624061
- Nagy, G., Presley, R., Jeon, H., …, **Gupta, A.**, Banerjee, T., Chung, D., Tsichlis, P. N., & Parvin, J. D. (2024). Multiplexed functional analysis of TAP2 variants in regulating MHC-I cell surface abundance reveals overexpression of PLK1 downregulates antigen presentation. *bioRxiv*. doi:10.1101/2024.09.30.615919

Manuscripts in Preparation

- **Gupta, A.**, et al. SRT: A statistical framework for the integrative analysis of spatial TCR-seq & spatial transcriptomics data. *In preparation.*
- **Gupta, A.**, et al. SCOPE: an interactive application for Bayesian stochastic block model clustering. *In preparation.*

PRESENTATIONS

- **Gupta, A.**, et al. (2026). *Joint Statistical Meetings (JSM)* 2026. Oral presentation: "SRT: A Statistical Framework for the Integrative Analysis of Spatial TCR-seq & Spatial Transcriptomics Data."
- **Gupta, A.**, et al. (2026). *International Conference on Frailty & Sarcopenia Research (ICFSR)* 2026, Washington, DC. Poster (presented by a co-author): Protein intake and lean soft tissue in a propensity-matched NHANES cohort.
- **Gupta, A.**, et al. (2026). *Midwest Biopharmaceutical Statistics Workshop (MBSW)* 2026. Poster: "SRT: A Statistical Framework for the Integrative Analysis of Spatial TCR-seq & Spatial Transcriptomics Data."
- **Gupta, A.**, et al. (2026). *ENAR Spring Meeting* 2026. Oral presentation: "SRT: A Statistical Framework for the Integrative Analysis of Spatial TCR-seq & Spatial Transcriptomics Data."
- **Gupta, A.**, et al. (2026). *Annual Joint Biostatistics Symposium* 2026, Columbus, OH. Poster: "SRT: A Statistical Framework for the Integrative Analysis of Spatial TCR-seq & Spatial Transcriptomics Data."
- **Gupta, A.**, et al. (2023). *ICSA Midwest Chapter & NIC-ASA Joint Fall Meeting* 2023. Poster: "Multi-disease genetic analysis with integration of functional annotation data."
- **Gupta, A.**, et al. (2023). *Annual Joint Biostatistics Symposium* 2023, Columbus, OH. Poster: "Multi-disease genetic analysis with integration of functional annotation data."

Departmental & Lab Talks

- Student seminar talks (4), Department of Statistics, The Ohio State University
- Research presentations at collaborating laboratories (multiple)
- Training talk on graph neural networks (GNNs), Biomedical Informatics Shared Resource (BISR), The Ohio State University Comprehensive Cancer Center, James (2025)

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA); Eastern North American Region of the International Biometric Society (ENAR); Institute of Mathematical Statistics (IMS); American Public Health Association (APHA)

AWARDS & SCHOLARSHIPS

- Graduate Career Accelerator Fund (CAF) Award, The Ohio State University (May 2026)
- MBSW Student Travel Grant (May 2026)
- Koch Graduate Student Travel Award, The Ohio State University (Mar 2026)
- ICSA Student Travel Grant (Oct 2023)
- INSPIRE SHE Scholarship, Department of Science & Technology, Govt. of India (2014 - 2019) Five-year award for students in the top 1 percentile

RESEARCH GROUP ACTIVITIES

- **Immuno-Oncology Informatics Group (IOIG), Pelotonia Institute** (2024 - 2025)
  Contributed to the institute's immuno-oncology research services through data analysis, statistical consultation, and training. Presented project updates at monthly group meetings
- **Chung Research Group, The Ohio State University**
  Regularly presented at lab meetings, giving detailed updates on research progress, reviewing relevant literature, and leading discussions among lab members

LEADERSHIP & SERVICE

- **Student Representative, Columbus Chapter, American Statistical Association (ASA)** (2026 - 2027)
- **President, Statistics & Biostatistics Graduate Student Association, The Ohio State University** (Aug 2024 - Aug 2025)
  Led a 100+ member organization. Liaised between graduate students and department administration
- **Officer, Public Health Graduate Student Association (PHGSA), The Ohio State University** (Aug 2022 - Aug 2023)
  Served as an officer of the graduate student association for the College of Public Health
- **Placement Coordinator, Department of Statistics, University of Delhi** (Jan 2018 - Jun 2019)
  Coordinated full-time placements for 70+ students and internships for 70+ students. Invited, hosted, and communicated with recruiting companies, and helped train students for the recruitment process
- **Judge, Statistical Analysis Award, State Science Day, Ohio Academy of Science** (2026)
  Evaluated student science-fair projects for the award sponsored by the Department of Statistics, The Ohio State University
- **Head Evaluator, FRAME Summer Fellowship, West Bengal**
  Led the evaluation of applications for FRAME (Fellowship for Research, Academic Mentorship and Exploration)

TEACHING EXPERIENCE

The Ohio State University **Columbus, OH, USA**
Instructor *Aug 2026 - Present*
Division of Biostatistics, College of Public Health
- PUBHBIO 2210: Biostatistics for Public Health Research (undergraduate)
  Teaching two sections (50 students in total)

The Ohio State University **Columbus, OH, USA**
Graduate Teaching Associate *Aug 2026 - Present*
Division of Biostatistics, College of Public Health
- PUBHBIO 7255: Introduction to Causal Inference (graduate)

The Ohio State University **Columbus, OH, USA**
Graduate Teaching Associate *Aug 2021 - May 2022*
Division of Biostatistics, College of Public Health
- PUBHBIO 2210: Biostatistics for Public Health Research (undergraduate)
- PUBHBIO 6270: Introduction to SAS for Public Health Students (graduate)

MENTORING

- **Graduate admissions mentoring** (2022 - Present)
  Mentored six students through the PhD admissions process and advised others applying to graduate programs. Currently mentoring a student from Denison University applying to PhD programs
- **Statistical mentoring for graduate researchers**
  Guided 3 graduate researchers on statistical concepts and on designing and carrying out the analyses for their research: an MDS candidate in oral and maxillofacial surgery (through her thesis defense), and PhD students in epidemiology and computational linguistics
- **Mentor, DecodingCOVID Virtual Hackathon, South Asian Medical Students' Association** (2020)

SOFTWARE

- **SCOPE:** R Shiny app for Bayesian stochastic block model clustering with posterior uncertainty quantification; used by 5+ labs in the USA, Italy, and the UK. chunglab.bmi.osumc.edu/SCOPE
- **SRT:** Statistical framework for integrating spatial TCR-seq and transcriptomics data

SKILLS

- **Clinical & Regulatory:** CDISC (SDTM, ADaM, TFL), Phase I-III trial programming, survival analysis, causal inference, mixed models
- **Omics & Genomics:** Seurat, DESeq2, Cell Ranger & Space Ranger (10x Genomics), PLINK (GWAS), SingleR, UCell, ScType, spatial transcriptomics, TCR-seq, DNA methylation
- **Statistics & Machine Learning:** Bayesian inference, MCMC, FDR control, deep learning, feature selection, Gaussian mixture models
- **Programming & Software:** R (Shiny, R Markdown, Quarto), SAS, LaTeX, VS Code; also C, C++, Stata, SPSS, SQL

VOLUNTEER WORK

- **Volunteer, Florence Nightingale Day for Statistics and Data Science, The Ohio State University** (2023)
  Supported an outreach event introducing middle- and high-school students to statistics and data science, held in partnership with the American Statistical Association and the Caucus for Women in Statistics
- **Volunteer, National Service Scheme (NSS)** (2016)
  Organized games and food stalls at Shishu Mela 2016, a recreational fair for underprivileged children who traveled from villages across West Bengal
- **Core Committee Member, Publication Committee, *Prakarsho*** (2016)
  Helped produce the annual magazine of the Department of Statistics, St. Xavier's College, Kolkata

REFERENCES

Available upon request.